
Chapter 7: Fractions

Worksheet 7: Types of Fractions and Equivalent Fractions

Learning Objective: Students will classify fractions (proper, improper, mixed); find equivalent fractions; simplify fractions; compare fractions using LCM; and convert between mixed numbers and improper fractions.

Worked Example

Convert $3\frac{4}{5}$ to an improper fraction and vice versa.

Solution: $3\frac{4}{5} = \frac{3 \times 5 + 4}{5} = \frac{19}{5}$

Convert $\frac{23}{7}$: $23 \div 7 = 3$ remainder 2; so $\frac{23}{7} = 3\frac{2}{7}$

Questions 1–5: Easy / Recall

Q1. Classify each fraction (proper, improper, or mixed): $\frac{3}{7}$; $\frac{9}{4}$; $2\frac{1}{3}$; $\frac{5}{5}$.

Q2. Convert $4\frac{2}{3}$ to an improper fraction.

Q3. Convert $\frac{31}{8}$ to a mixed number.

Q4. Write two equivalent fractions for $\frac{3}{5}$.

Q5. Reduce $\frac{36}{48}$ to its simplest form.

Questions 6–10: Basic Application

Q6. Compare $\frac{5}{8}$ and $\frac{7}{12}$. Which is greater?

Q7. Arrange in descending order: $\frac{2}{3}$, $\frac{5}{6}$, $\frac{3}{4}$, $\frac{7}{12}$.

Q8. A class has 40 students. $\frac{3}{8}$ are girls. How many are boys?

Q9. Simplify: $\frac{72}{108}$.

Q10. Find a fraction between $\frac{1}{4}$ and $\frac{1}{2}$.

Questions 11–15: Practice and Skill Building

Q11. $\frac{5}{6} + \frac{7}{8} =$

Q12. $3\frac{1}{4} - 1\frac{5}{6} =$

Q13. $\frac{4}{5} \times \frac{15}{16} =$

Q14. $\frac{7}{8} \div \frac{21}{32} =$

Q15. The product of two fractions is $\frac{5}{12}$. One fraction is $\frac{5}{8}$. Find the other.

Questions 16–18: Higher Order Thinking

Q16. Raju spends $\frac{1}{4}$ of his money on food, $\frac{1}{3}$ on rent, and $\frac{1}{6}$ on travel. He saves the rest. If he saves 1500, find his total income.

Q17. A tank is $\frac{3}{5}$ full. After adding 45 litres, it is $\frac{4}{5}$ full. Find the capacity of the tank.

Q18. A recipe for 6 people needs $2\frac{1}{4}$ cups of flour. How much flour is needed for 10 people? For 4 people?

Questions 19–20: Real-Life Applications

Q19. In a school of 1200 students, $\frac{5}{12}$ play cricket, $\frac{1}{4}$ play football, and $\frac{1}{6}$ play badminton. How many play each sport? How many play none of these?

Q20. A shopkeeper had 60 kg of rice. He sold $\frac{3}{5}$ on Monday and $\frac{1}{4}$ of the remainder on Tuesday. How much was sold on Tuesday? How much is left?

Reflection Question: Explain the difference between dividing a fraction by a whole number and multiplying a fraction by a whole number. Give examples.